Ch. 17

(37.) Lightning strike: $\Delta t = 45\ ms = 45 \times 10^{-3}\ sec$
$\Delta Q = 350\ C$

$\rightarrow I = \frac{\Delta Q}{\Delta t} = \frac{350\ C}{45 \times 10^{-3}\ s} = \boxed{7.78 \times 10^{3}\ amp}$

(41.) Platinum, room temp resistance $R_0 = 58.5\ \Omega$
$\rho = 1.06 \times 10^{-7}\ \Omega \cdot m$ [resistivity]
$\alpha = 3.9 \times 10^{-3}\ \frac{1}{^\circ C}$ [temp coefficient]

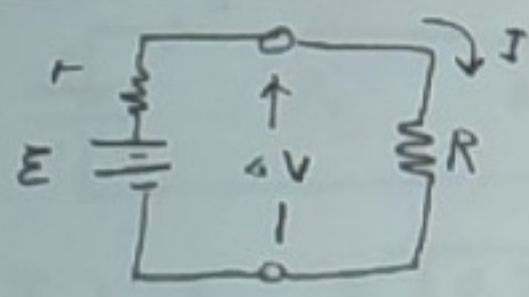
$\rho = \rho_0 [1 + \alpha (T - T_0)]$

$\rightarrow R = R_0 [1 + \alpha (T - T_0)]$

$\Delta T = T - T_0 = 0.1^\circ C$

$\Delta R = R - R_0$
$= R_0 \alpha (T - T_0)$
$= R_0 \alpha \Delta T$
$= \boxed{0.0228\ \Omega}$

(47.)



$\varepsilon = 18$ volts
$\Delta V = 17.4$ volts
$R = 130\ \Omega$

Determine internal resistance r

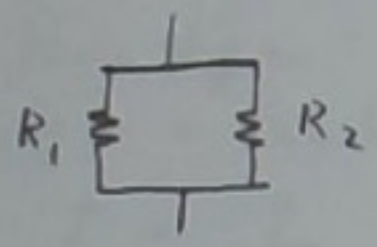
$0 = \varepsilon - Ir - IR$
$= \varepsilon - I(r + R)$

$I = \frac{\Delta V}{R} = \boxed{0.134\ Amp}$

Thus, $\varepsilon = I(r + R)$
$\rightarrow r + R = \varepsilon / I$

$r = \varepsilon / I - R = \frac{18\ volts}{0.134\ A} - 130\ \Omega = \boxed{4.48\ \Omega}$

(53.)



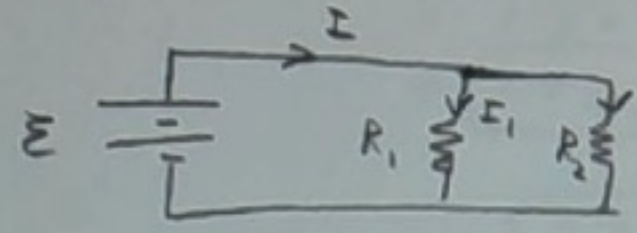
$\equiv$



$R_1 = 250\,\Omega$
$R_2 = 370\,\Omega$

a) $\frac{1}{R_p} = \frac{1}{R_1} + \frac{1}{R_2} \rightarrow R_p = \frac{1}{\left(\frac{1}{R_1} + \frac{1}{R_2}\right)} = \boxed{149\,\Omega}$

b)



$\mathcal{E} = 12\ V$

$I = \frac{\mathcal{E}}{R_p} = \frac{12\,V}{149\,\Omega} = \boxed{0.080\ A}$

Note that $I = I_1 + I_2$

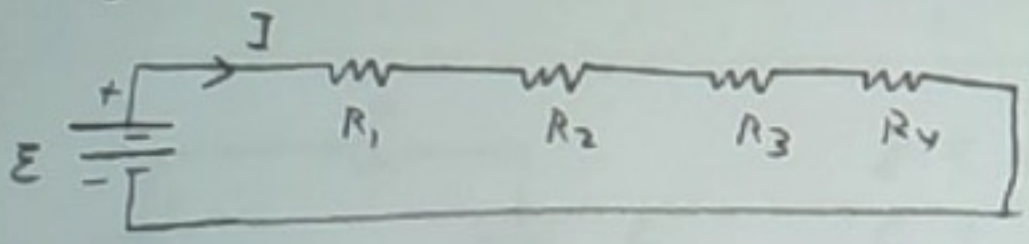
$I_1 = \frac{\mathcal{E}}{R_1} = \frac{12\,V}{250\,\Omega} = \boxed{0.048\ A}$

$I_2 = \frac{\mathcal{E}}{R_2} = \frac{12\,V}{370} = \boxed{0.032\ A}$

(55.) $R_1 = 12\,\Omega$, $R_2 = 15\,\Omega$, $R_3 = 20\,\Omega$, $R_4 = 35\,\Omega$

a) $R_{eq} = R_1 + R_2 + R_3 + R_4 = \boxed{82\,\Omega}$ (series combination)

b)



, $\mathcal{E} = 6.0$ volts

$I = \frac{\mathcal{E}}{R_{eq}} = \boxed{0.73\ amp}$

$\Delta V_1 = R_1 I = \boxed{0.88\ volt}$

$\Delta V_2 = R_2 I = \boxed{1.10\ volts}$

$\Delta V_3 = R_3 I = \boxed{1.46\ volts}$

$\Delta V_4 = R_4 I = \boxed{2.56\ volts}$

Note: $\Delta V_1 + \Delta V_2 + \Delta V_3 + \Delta V_4 = 6.0\ volt = \mathcal{E}$

(69.) Hair dryer: $P = 1750$ watts

$\Delta V = 120\ V$

a) $I = \frac{P}{\Delta V} = \boxed{14.58\ \text{Amps}}$

b) Energy: $E = Pat$, $\Delta t = 10\ \text{min} = 600\ s$

$= 1750\ W \times 600\ s$

$= \boxed{1.05 \times 10^6\ J} \times \left(\frac{1\ \text{kW.h}}{3.6 \times 10^6\ \text{Joules}} \right)$

$= \boxed{0.292\ \text{kWh}}$

c) cost per kWh: $c = \frac{15\ ¢}{\text{kW.h}}$

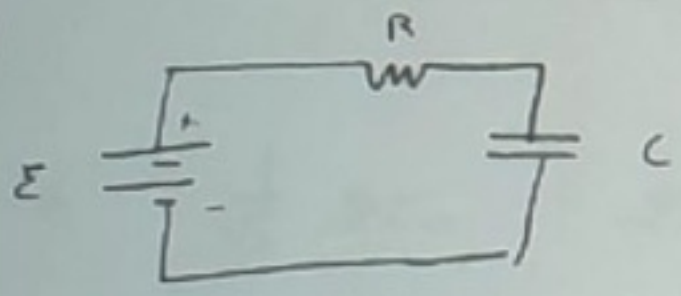
Monthly cost:

$\text{monthly cost} = \frac{15\ ¢}{\text{kWh}} \times \left(\frac{0.292\ \text{kWh}}{\text{day}} \right) \times 30\ \text{day}$

$= 131\ ¢$

$= \boxed{\$1.31}$

(75.)



$\mathcal{E} = 6\ \text{volts}$

$C = 300\ \text{pF} = 300 \times 10^{-12}\ F$

$R = 175\ \text{M}\Omega = 175 \times 10^6\ \Omega$

$q(t) = q_{max}(1 - e^{-t/\tau})$

where $q_{max} = C \cdot \mathcal{E} = \boxed{1.8\ \text{nC}}$, $\tau = RC = 0.0525\ s$.

a) $q(t = 1\ \text{ms}) = q_{max}(1 - e^{-1 \times 10^{-3}\ s/\tau}) = \boxed{0.034\ \text{nC}}$

b) $q(t = 10\ \text{ms}) = q_{max}(1 - e^{-10 \times 10^{-3}\ s/\tau}) = \boxed{0.312\ \text{nC}}$

c) $q(t = 100\ \text{ms}) = q_{max}(1 - e^{-100 \times 10^{-3}\ s/\tau}) = \boxed{1.53\ \text{nC}}$

(11.) $q(t) = q_{max}(1 - e^{-t/\tau})$

$\tau = RC, \quad q_{max} = \varepsilon C$

a) what is t/τ for $q(t) = \frac{1}{2} q_{max}$?

Ans: $\frac{1}{2} q_{max} = q_{max}(1 - e^{-t/\tau})$

$\frac{1}{2} = 1 - e^{-t/\tau}$

$e^{-t/\tau} = 1 - \frac{1}{2} = \frac{1}{2}$

$\rightarrow \quad -t/\tau = \ln(\frac{1}{2})$

$t/\tau = \ln(2) = \boxed{0.693}$

b) what is t/τ for $U(t) = \frac{1}{2} U_{max}$?

Recall: $U_{max} = \frac{1}{2} q_{max} \varepsilon = \frac{1}{2C} q_{max}^2$

$U(t) = \frac{1}{2C} q^2(t)$

i.e., $U(t) = \frac{1}{2} U_{max} \iff q^2(t) = \frac{1}{2} q_{max}^2$

$q(t) = \frac{1}{\sqrt{2}} q_{max}$

$\rightarrow \quad \frac{1}{\sqrt{2}} q_{max} = q_{max}(1 - e^{-t/\tau})$

$\frac{1}{\sqrt{2}} = 1 - e^{-t/\tau}$

$e^{-t/\tau} = 1 - \frac{1}{\sqrt{2}}$

$\rightarrow \quad -t/\tau = \ln(1 - \frac{1}{\sqrt{2}})$

$t/\tau = -\ln(1 - \frac{1}{\sqrt{2}}) = \boxed{1.23}$